# Aviv Shaked

Cybersecurity Developer & Researcher · 18 · Haifa, Israel · Open to remote / on-site / hybrid

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## SUMMARY

**C++20 cybersecurity developer and researcher**, 18, with deep expertise in network protocols, covert channels, binary exploitation, and reverse engineering. Graduate of **Magshimim+**, Israel's elite, selectively admitted national cybersecurity track. Architect of **HitchHiker**, a production-grade network steganography platform with zero observable network presence, built end-to-end in C++20 with a full applied cryptography stack. Placed **2nd globally** in the AI4Y World Competition (2023). HackTheBox (RE, Crypto, Pwn).

## TECHNICAL SKILLS

| | |
|---|---|
| Languages | C++20 (primary, 3+ years), Python, x86/x86-64 Assembly, 8086 Assembly, C, SQL, C# |
| C++ & Systems | RAII, smart pointers, multi-threading, modern C++20 idioms, CMake, interface abstraction, modular builds |
| Networking | Full network stack (L2–L7): TCP/IP, UDP, DNS, TLS, ICMP, ARP, DHCP · header-level field manipulation · kernel-level packet interception · iptables · Wireshark · Scapy |
| Security | Binary exploitation · ROP chains · buffer overflows · ASLR bypass · reverse engineering · network steganography · protocol analysis · threat modeling · exploit development · HackTheBox (RE / Crypto / Pwn) |
| Cryptography | Asymmetric & symmetric encryption · key exchange · key derivation · password hashing & salting · cryptanalysis · hash attacks · applied in production (Crypto++) |
| AI / ML | Neural network fundamentals · LLM/API integration · Python AI/ML tooling (PyTorch, TensorFlow, Scikit-learn) |
| Tools | Binary Ninja · IDA Pro · GDB · nmap · Git · Kali Linux · Docker · AWS EC2 · OpenVPN · JetBrains |

## PROJECTS

### HitchHiker: Network Steganography Platform

C++20 · 2025–2026 · Completed

Co-developed with Daniel Varkin · Repo private, architecture overview at **patcholie.github.io**

C++20 · Crypto++ · NFQueue · FTXUI · Redis · SQLite · Docker · AWS EC2 · CMake

- Architected a covert communication platform that embeds encrypted messages into live network traffic: **zero new packets generated**, no observable network presence, undetectable by design
- Full applied cryptography stack (key exchange, symmetric encryption, key derivation, password hashing) applied at bit-field precision across multiple protocol layers via kernel-level packet interception
- **Pluggable carrier architecture** behind a unified HitchHikerAPI; three independent CMake targets (lib / server / client), Redis + SQLite storage, full TUI client (FTXUI); server deployed on AWS EC2, clients run on Linux via Docker

### Multiplayer Trivia Game System

C++ / C# / Unity · Magshimim · 2024

C++ · C# · Unity · SQLite · TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server, C# Unity client, and SQLite database integrated end-to-end with clean OOP architecture

### Analisis: AI Wound Treatment Platform

Python · 2023 · Public (AGPL v3.0)

- Co-led development of an AI-powered wound diagnosis and treatment guidance platform; **placed 2nd globally** in the AI4Y World Competition (2023) across international teams

### Mancala-AI

Python · PyTorch · 2025–2026

- Multi-agent AI framework with multiple strategy types (including neural network agents); full training and tournament evaluation pipeline with head-to-head agent benchmarking

## EDUCATION

### Magshimim Cyber, Advanced Track (Magshimim+)

2023–2026

Israel's elite national cybersecurity program, selective advanced track · networking, binary exploitation, reverse engineering, cryptography, systems programming (C/C++), Linux, and team-based security development

### BOSMAT, Haifa

2020–2026

Computer Science + AI · 10 study units (maximum track)

## EXPERIENCE

### Tools Developer, AllBikes

2022–2023 · Remote

- Built Python automation tools for web scraping and product catalog structuring; **tooling adopted by the full team**, delivered independently in a remote professional environment while in secondary school

## AWARDS & RECOGNITION

**2nd Place, AI4Y World Competition** · 2023, Global **Magshimim+ Advanced Track** · Graduate 2026 **Magshimim Cyber** · Graduate 2026